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DATA STRUCTURE AND ALGORITHMS

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Data Structures and Algorithms

In the most general sense, a data structure is any data representation and its associated operations. Even an integer or floating point number stored on the computer can be viewed as a simple data structure. More typically, a data structure is meant to be an organization or structuring for a collection of data items. A sorted list of integers stored in an array is an example of such a structuring. An algorithm is a step-by-step process designed to solve a specific problem or reach a goal. A recipe for french fries is an everyday example: it outlines the exact instructions needed to prepare a meal. In computer science, algorithms follow the same logic, but the steps are written in code and process data structures instead of ingredients. As a core foundation of programming, a well-designed algorithm not only solves the target problem but can dramatically improve a program's speed and performance.

Linear Data Structure

Linear Data Structures are a type of data structure in computer science where data elements are arranged sequentially or linearly. Each element has a previous and next adjacent, except for the first and last elements.

One-Dimension Array

This is the simplest form of an array, which consists of a single row of elements, all of the same data type. Elements in a 1D array are accessed using a single index.



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The Big O Notation

Big O notation measures an algorithm's time or space complexity by describing its asymptotic behavior—how its execution time or memory requirements grow as the input size scales—rather than calculating an exact value. By focusing primarily on the worst-case scenario, Big O establishes a strict theoretical upper bound on resource usage. This makes it an essential tool in computer science for comparing the relative efficiency of different algorithms and data structures as inputs become arbitrarily large.

Basic Searching Algorithms

Searching algorithms are essential tools in computer science used to locate specific items within a collection of data. In this tutorial, we are mainly going to focus upon searching in an array. When we search an item in an array, there are two most common algorithms used based on the type of input array.

Linear Search

In Linear Search, we iterate over all the elements of the array and check if the current element is equal to the target element. If we find any element to be equal to the target element, then return the index of the current element. Otherwise, if no element is equal to the target element, then return -1 as the element is not found. Linear search is also known as sequential search.

Binary Search

It is a searching algorithm that operates on a sorted or monotonic search space, repeatedly dividing it into halves to find a target value or optimal answer in logarithmic time $O(\log N)$.



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